

Exam Date & Time: 25-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024

Computer Vision [IT-4031]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Examine 2D Convolution Filter in detail. Solve the given input image (f) with its kernel (h) using convolution filtering method and display the output image (g) for the pixel values - (0, 1), (1, 2), (2, 3), (3, 4).

A)

Input Image (f)					Kernel (h)		
1	0	1	2	2	0	1	0
1	1	2	2	3	1	-4	1
1	2	2	6	3	0	1	0
1	1	2	2	3			

(5)

- B) Justify why filters are used on an image. Also, discuss various filters used for noise reduction on an image. (3)
- C) Define Computer Vision? Why is vision so difficult? Discuss various color models that are involved in computer vision. (2)
- 2) With respect to Scale Invariant Feature Transform (SIFT), examine the following steps to extract and interpret the key points in an image. (5)
- (i) Scale space peak selection.
- A) (ii) Key point localization. (5)
- (iii) Orientation assignment.
- (iv) Key point descriptor.
- B) Identify various image feature detection methods and discuss in detail. (3)
- C) How is the detection of humans achieved through the utilization of Histogram of Oriented Gradient (HOG) features? (2)
- 3) Examine the mathematics involved in computing the 3D coordinates of a point illuminated by a light stripe plane having equation $aX + bY + cZ + d = 0$ using a known camera calibration matrix C. (5)
- A)
- B) What is the reason behind tilting of Epipolar lines? How do you correct this to get a horizontal line to simplify the search for corresponding points between two images? (5)

Demonstrate the necessary transformation matrix.

- 4) Differentiate the distinction between motion field and optical flow in an image. Use a diagram to illustrate your answer. (5)

A)

- B) What are Support Vector Machines? Demonstrate with an examples and neat illustrations and list out the advantages and disadvantages of SVM. (5)

- 5) Organize the steps involved in tracking of features using KLT method. (5)

A)

- B) For the following given three text sentences:

S1 -> This movie is very scary and long.

S2 -> This movie is not scary and is slow.

S3 -> This movie is spooky and good. (5)

Apply Bag of words method for given three text sentences and also demonstrate Term Frequency - Inverse-Document Frequency (TF-IDF) for the given same text sentences.

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